iSpark Attendance API

0.1.0 OAS 3.1

/openapi.json

API for managing cohorts, trainers, and student attendance.

default

GET / Test Database Connection

Test endpoint to verify our database connection is working inside FastAPI.

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header.	*No links*

Code	Description	Links

Example Value Schema

```
"string"
```

GET `/programs` List Programs

Get a list of all Programs.

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*

Code	Description	Links


```
[
  {
    "name": "string",
    "description": "string",
    "id": 0,
    "created_at": "2026-10-15T14:00:28.611Z"
  }
]
```


POST /programs Create Program

Create a new Program.

- We expect 'name' and 'description' in the JSON body.
- Pydantic ensures 'name' is not empty and is under 150 characters.

Parameters

Try it out

No parameters

Request body required

application/json

Example Value Schema


```
{
  "name": "string",
  "description": "string"
}
```


Responses

Code	Description	Links
201	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema `{ "name": "string", "description": "string", "id": 0, "created_at": "2026-10-15T14:00:28.613Z" }`	*No links*
422	Validation Error Media type **application/json** **Example Value** Schema `{ "detail": [ { "loc": [ "string", 0 ], "msg": "string", "type": "string", "input": "string", "ctx": {} } ] }`	*No links*

POST `/cohorts` Create Cohort

Create a new Cohort under a specific Program.

Parameters Try it out

No parameters

Request body required application/json

Example Value Schema


```
{
  "program_id": 0,
  "name": "string",
  "start_date": "2026-10-15",
  "end_date": "2026-10-15"
}
```


Responses

Code	Description	Links
201	Successful Response Media type application/json Controls `Accept` header. **Example Value** Schema	*No links*


```
{
  "program_id": 0,
```

Code	Description	Links


```
  "name": "string",
  "start_date": "2026-10-15",
  "end_date": "2026-10-15",
  "id": 0,
  "created_at": "2026-10-15T14:00:28.615Z"
}
```


422 Validation Error — *No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET `/programs/{program_id}/cohorts` List Cohorts For Program

List all cohorts belonging to a specific program.

Parameters Try it out

Name	Description
program_id * required integer *(path)*	program_id

Responses

Code	Description	Links

200	Successful Response
	
	Media type
	application/json
	Controls `Accept` header.
	
	Example Value Schema
	
	[
	  {
	  "program_id": 0,
	  "name": "string",
	  "start_date": "2026-10-15",
	  "end_date": "2026-10-15",
	  "id": 0,
	  "created_at": "2026-10-15T14:00:28.617Z"
	  }
	]
		No links

422	Validation Error Media type **application/json** **Example Value** Schema	*No links*

Code	Description	Links


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET /cohorts/{cohort_id} Get Cohort

View details of a single cohort.

Parameters

Try it out

Name	Description
cohort_id * required integer *(path)*	cohort_id

Responses

Code	Description	Links
200	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema `{ "program_id": 0, "name": "string", "start_date": "2026-10-15", "end_date": "2026-10-15", "id": 0, "created_at": "2026-10-15T14:00:28.619Z" }`	*No links*
422	Validation Error Media type **application/json** **Example Value** Schema `{ "detail": [ { "loc": [ "string", 0 ], "msg": "string", "type": "string", "input": "string", "ctx": {} } ] }`	*No links*

POST `/trainers` Create Trainer

Create a new trainer profile.

Parameters Try it out

No parameters

Request body required application/json

Example Value Schema

```
{
  "name": "string",
  "email": "user@example.com"
}
```

Responses

Code	Description	Links
201	Successful Response Media type application/json Controls `Accept` header. **Example Value** Schema	*No links*

Code	Description	Links


```
{
  "name": "string",
  "email": "user@example.com",
  "id": 0,
  "created_at": "2026-10-15T14:00:28.621Z"
}
```


422 Validation Error *No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


POST **/cohorts/{cohort_id}/trainers** Assign Trainer To Cohort

Assign an existing trainer to a cohort.

Parameters Try it out

Name	Description
cohort_id * required integer *(path)*	cohort_id

Request body required

application/json

Example Value Schema

```
{
  "trainer_id": 0
}
```

Responses

Code	Description	Links
201	Successful Response Media type application/json Controls `Accept` header. **Example Value** Schema `"string"`	*No links*

Code	Description	Links
422	Validation Error Media type **application/json** **Example Value** Schema	*No links*


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


POST `/trainees` Create Trainee

Create a new trainee profile.

Parameters Try it out

No parameters

Request body required application/json

Example Value Schema

```
{
  "name": "string",
  "email": "user@example.com"
}
```


Responses

Code	Description	Links
201	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema `{ "name": "string", "email": "user@example.com", "id": 0, "created_at": "2026-10-15T14:00:28.627Z" }`	*No links*
422	Validation Error Media type **application/json** **Example Value** Schema `{ "detail": [ { "loc": [ "string", 0 ],`	*No links*

Code	Description	Links


```
        "msg": "string",
        "type": "string",
        "input": "string",
        "ctx": {}
      }
    ]
}
```


POST /cohorts/{cohort_id}/enrollments Enroll Trainee In Cohort

Enroll an existing trainee into a cohort.

Parameters

Try it out

Name	Description
cohort_id * required integer *(path)*	cohort_id

Request body required

application/json

Example Value Schema


```
{
  "trainee_id": 0
}
```

Responses

Code	Description	Links
201	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema `"string"`	*No links*
422	Validation Error Media type **application/json** **Example Value** Schema `{ "detail": [ { "loc": [ "string", 0 ], "msg": "string", "type": "string", "input": "string", "ctx": {} } ] }`	*No links*

POST `/cohorts/{cohort_id}/classes` Schedule Class

Schedule a class for a cohort.

Parameters Try it out

Name	Description
cohort_id * required integer *(path)*	cohort_id

Request body required application/json

Example Value Schema

```
{
  "trainer_id": 0,
  "date": "2026-10-15",
  "topic": "string"
}
```

Responses

Code	Description	Links
201	Successful Response Media type	*No links*

Code	Description	Links

	application/json
	Controls `Accept` header.
	
	Example Value Schema
	
	{
	"trainer_id": 0,
	"date": "2026-10-15",
	"topic": "string",
	"id": 0,
	"cohort_id": 0,
	"created_at": "2026-10-15T14:00:28.635Z"
	}	

422	Validation Error
	
	Media type
	application/json
	
	Example Value Schema
	
	{
	"detail": [
	{
	"loc": [
	"string",
	0
	],
	"msg": "string",
	"type": "string",
	"input": "string",
	"ctx": {}
	}
	]
	}
		No links

GET **/cohorts/{cohort_id}/classes** List Cohort Classes

List a cohort's classes by date.

Parameters

Try it out

Name	Description
cohort_id * required `integer` *(path)*	cohort_id

Responses

Code	Description	Links
200	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*


```
[
  {
    "trainer_id": 0,
    "date": "2026-10-15",
    "topic": "string",
    "id": 0,
    "cohort_id": 0,
    "created_at": "2026-10-15T14:00:28.636Z"
  }
]
```

Code	Description	Links
422	Validation Error Media type **application/json** **Example Value** Schema	*No links*


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


POST `/classes/{class_id}/attendance` Record Attendance

Record OR Correct attendance for a class in a single submission.

Parameters Try it out

Name	Description
class_id * required integer *(path)*	class_id

Request body required

application/json

Example Value Schema


```
{
  "recorded_by_trainer_id": 0,
  "records": [
    {
      "trainee_id": 0,
      "status": "Present"
    }
  ]
}
```


Responses

Code	Description	Links
200	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema `"string"`	*No links*

Code	Description	Links
422	Validation Error Media type application/json **Example Value** Schema	*No links*


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET `/cohorts/{cohort_id}/report` Get Cohort Report

Requirement #8: Cohort Report. Shows every trainee, their status counts, attendance rate, and flags < 75%.

Parameters Cancel

Name	Description
cohort_id * required integer *(path)*	2

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/cohorts/2/report' \
  -H 'accept: */*'
```

Request URL

```
http://127.0.0.1:8000/cohorts/2/report
```

Server response

Code	Details
200	**Response body** [{ "trainee_id": 7, "trainee_name": "Student 7", "present_count": 0, "absent_count": 0, "late_count": 4, "excused_count": 0, "total_classes_marked": 4, "attendance_rate_percentage": 50,

Code	Details


```
    "is_flagged": true
  },
  {
    "trainee_id": 8,
    "trainee_name": "Student 8",
    "present_count": 0,
    "absent_count": 0,
    "late_count": 0,
    "excused_count": 4,
    "total_classes_marked": 4,
    "attendance_rate_percentage": 100,
    "is_flagged": false
  },
  {
    "trainee_id": 9,
    "trainee_name": "Student 9",
    "present_count": 4,
    "absent_count": 0,
```


Download

Response headers


```
content-length: 958
content-type: application/json
date: Thu,15 Oct 2026 11:52:41 GMT
server: uvicorn
```


Responses

Code	Description	Links
200	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*


```
[
  {
    "trainee_id": 0,
    "trainee_name": "string",
    "present_count": 0,
    "absent_count": 0,
    "late_count": 0,
    "excused_count": 0,
```

Code	Description	Links


```
    "total_classes_marked": 0,
    "attendance_rate_percentage": 0,
    "is_flagged": true
  }
]
```


422 Validation Error *No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET /cohorts/{cohort_id}/trainees/{trainee_id}/history Get Trainee History

Requirement #9: Trainee history for one cohort in date order.

Parameters Try it out

Name	Description
cohort_id * required integer *(path)*	cohort_id
trainee_id * required integer *(path)*	trainee_id

Responses

Code	Description	Links

200	Successful Response
	
	Media type
	application/json
	Controls `Accept` header.
	
	Example Value Schema
	
	[
	{
	"class_id": 0,
	"date": "2026-10-15",
	"topic": "string",
	"status": "Present"
	}
	]
		No links

422	Validation Error Media type	*No links*

Code	Description	Links

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET `/classes/{class_id}/summary` Get Class Summary

Requirement #10: Class summary (status counts and unmarked trainees).

Parameters Try it out

Name	Description
class_id * required `integer` *(path)*	class_id

Responses

Code	Description	Links

200	Successful Response
	
	Media type
	application/json
	Controls `Accept` header.
	
	Example Value Schema
	
	{
	"class_id": 0,
	"status_counts": {
	"additionalProp1": 0,
	"additionalProp2": 0,
	"additionalProp3": 0
	},
	"unmarked_trainees": [
	"string"
	]
	}
		No links

422	Validation Error
	
	Media type
	application/json
	
	Example Value Schema
	
	{
	"detail": [
	{
	"loc": [
	"string",
	0
	],
	"msg": "string",
	"type": "string",
	"input": "string",
	"ctx": {}
	}
		No links

Code	Description	Links


```
  ]
}
```


Schemas

AttendanceRecordCreate › Expand all `object`

AttendanceStatus › Expand all `string`

BulkAttendanceSubmit › Expand all `object`

ClassCreate › Expand all `object`

ClassResponse › Expand all `object`

ClassSummary › Expand all `object`

CohortCreate › Expand all `object`

CohortResponse › Expand all `object`

HTTPValidationError › Expand all `object`

ProgramCreate › Expand all `object`

ProgramResponse › Expand all `object`

TraineeCreate › Expand all `object`

TraineeEnroll › Expand all `object`

TraineeHistoryRecord › Expand all `object`

TraineeReport › Expand all `object`

TraineeResponse › Expand all `object`

TrainerAssign › Expand all `object`

TrainerCreate › Expand all `object`

TrainerResponse › Expand all `object`

ValidationError › Expand all `object`